

NET731 - Cloud Networking

Final Presentation

Slide 1

TechBridge Solutions Azure Infrastructure

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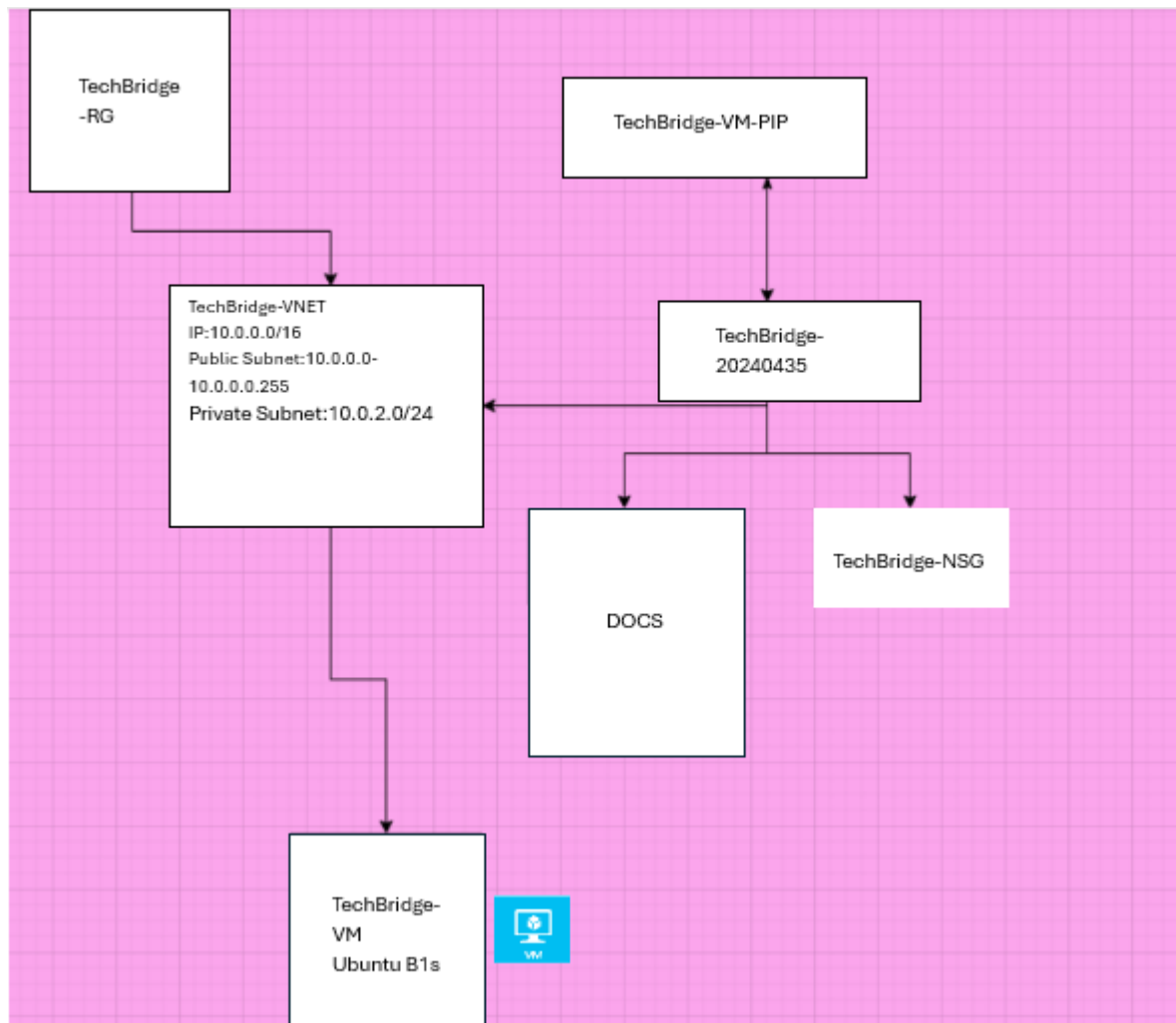
May 2026

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Project Overview

- Client: TechBridge Solutions (small business)
- Goal: Secure cloud infrastructure with web & internal tiers
- Platform: Microsoft Azure (10 weeks)
- Outcome: Complete environment - VNet, VM, NSG, Entra ID, Storage, Monitoring, Cost controls

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Key Components:

- TechBridge-RG - Resource group
- TechBridge-VNet - 10.0.0.0/16
- Public-Subnet - 10.0.1.0/24 | Private-Subnet - 10.0.2.0/24
- TechBridge-VM - Ubuntu B1s
- TechBridge-NSG - HTTP/HTTPS/SSH(MyIP)/Deny All
- techbridge20240435 - Storage (LRS)

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Decision	What I Chose	Why	Trade-off
VM Size	B1s (1 vCPU, 1GB RAM)	Cheapest, perfect for learning	Not for production
NSG Rule	Explicit Deny at priority 1000	Clear security intent	One extra rule
Storage	LRS replication	Most cost-effective	No DR protection

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SECURITY

NSG Inbound Rules:

- HTTP (80) - Any
- HTTPS (443) - Any
- SSH (22) - My IP only
- Deny All (Any)

RBAC (Least Privilege):

- Dev → Contributor
- Ops → Reader
- Finance → Reader

Defender Remediations:

- Closed SSH to internet
- Enabled OS updates

MONITORING

Azure Monitor:

- VM Insights enabled

NSG diagnostics to Log Analytics

Metrics:

- CPU Percentage
- Network In/Out
- Disk Read/Write

Alert:

- CPU > 80% for 5 min
- Email to 20240435

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Budget: \$100/month

Alerts: 80% (\$80) and 100% (\$100)

Actual Spend:

- VM (B1s): ~\$15/month
- Storage (LRS): ~\$1.50/month
- Public IP: ~\$3.00/month
- TOTAL: ~\$19.50/month

Savings:

Auto-shutdown at 7pm

B1s size

LRS replication

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Challenge 1:

B1s not available in my region

Solution: Switched to East US region

Lesson: Check availability first

Challenge 2:

NSG SSH rule not blocking traffic

Solution: Changed priority (lower number = higher priority)

Lesson: Priority 100 before priority 1000

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What I'd Do Differently:

Plan naming conventions before Week 2

Take screenshots immediately after each config

Set up budget alerts on Day 1

New Skills:

Azure Portal navigation

CIDR notation & subnet planning

RBAC & least privilege

NSG prioritization

Azure Monitor & alerts

Cost analysis

Advice:

Start GitHub early, commit often

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VNet with public/private subnets

B1s Ubuntu VM with static IP

Entra ID - 6 users, 3 groups, RBAC

NSG - HTTP/HTTPS/SSH(MyIP)/Deny All

Storage Account with SAS tokens

Azure Monitor with CPU alert

Cost management - \$45 spent of \$100

Key Takeaway:

Plan first, then build. Security, monitoring,
and cost must be considered from Day 1.

Thank You!

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